Theme- Open Innovation

VisionCam

AI Driver Alert System

By Team Neptune

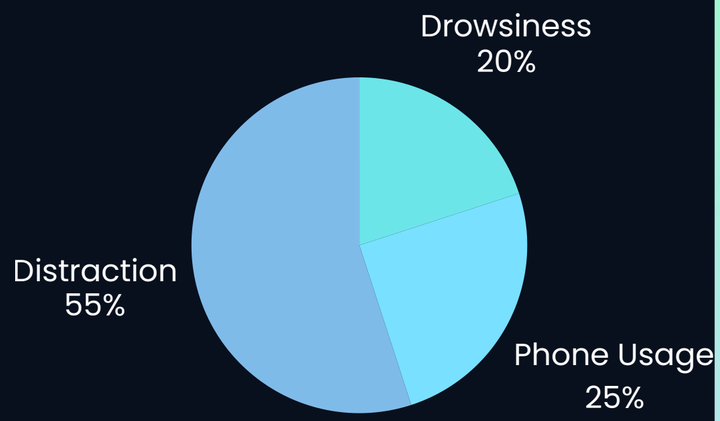
PROBLEM STATEMENT

"A second of distraction, and a life could be lost."

Problem: Every year, thousands of accidents occur due to driver drowsiness and mobile phone usage, causing severe injuries and fatalities worldwide.

Gaps in Current Solutions

- Existing driver monitoring systems are hardware-based and expensive (₹30,000–₹1,50,000).
- Limited accessibility — mostly used in premium cars or large commercial fleets.
- Most solutions detect only drowsiness, not distraction or phone usage.
- No camera-based system that works using a normal phone or laptop
- Lack of real-time alerts— drivers often realize fatigue only after it becomes risky



Why It Matters

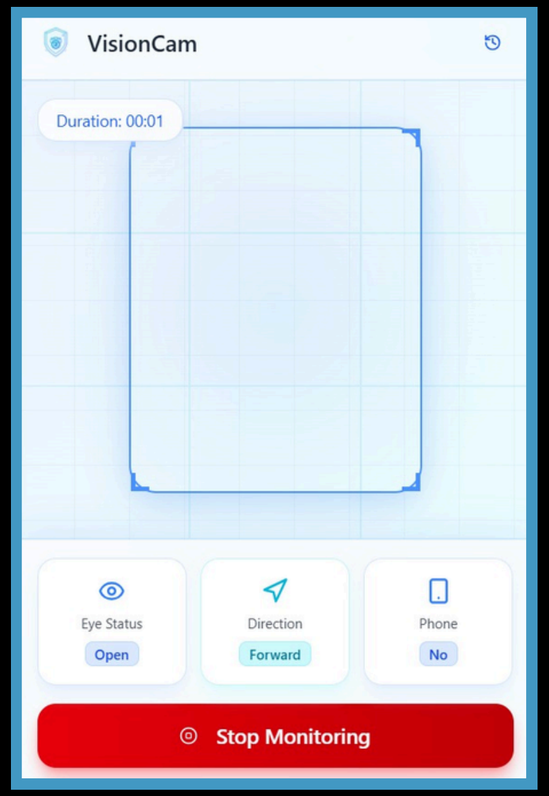
- 20% of road accidents are caused by driver fatigue or distraction (WHO Report).
- Even a split-second of inattention can lead to irreversible consequences.
- Manual monitoring or self-awareness isn't reliable during long drives.
- Accidents cause loss of life, property damage, and major economic costs.
- There's a pressing need for an affordable, AI-powered preventive solution.

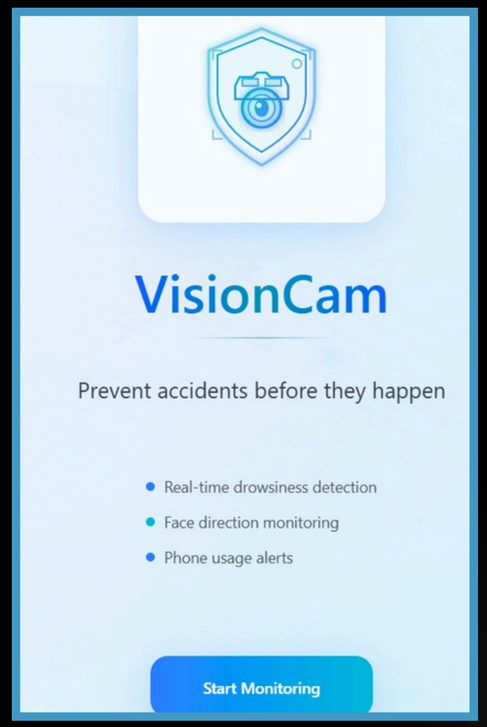
SOLUTION

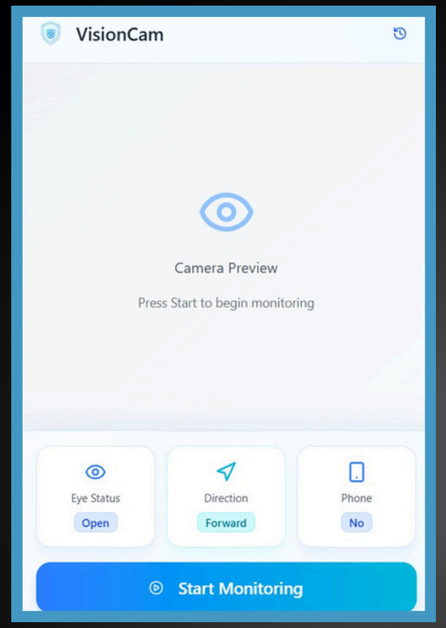
"An AI-powered co-driver that never sleeps — detecting fatigue, distraction, and danger in real time."

How It Works:

- Image Capture: Continuously captures the driver's face using the front camera
- AI Detection: Deep learning model analyzes blink rate, eye closure duration, gaze direction, and hand movement to detect signs of drowsiness or phone use.
- Alert System: If the system detects fatigue or distraction, it activates a beep or voice alert to regain the driver's attention







TECH STACK & ARCHITECTURE

FrontEnd-

- HTML / CSS / JavaScript
- Webcam API (live video feed)

AI Layer-

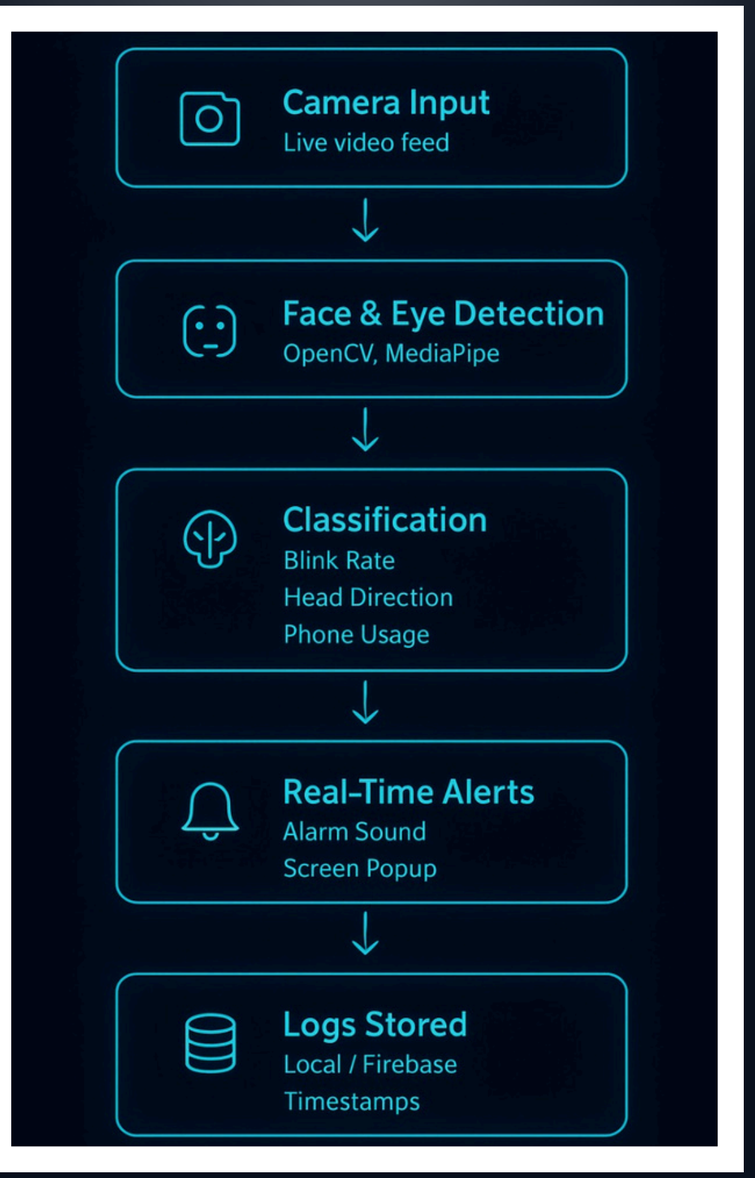
- OpenCV / MediaPipe (face + eye tracking)
- Custom alert logic (eye blink, head turn, mobile detection)

Backend-

- Firebase / Firestore DB
- Node.js API (optional for cloud storage)

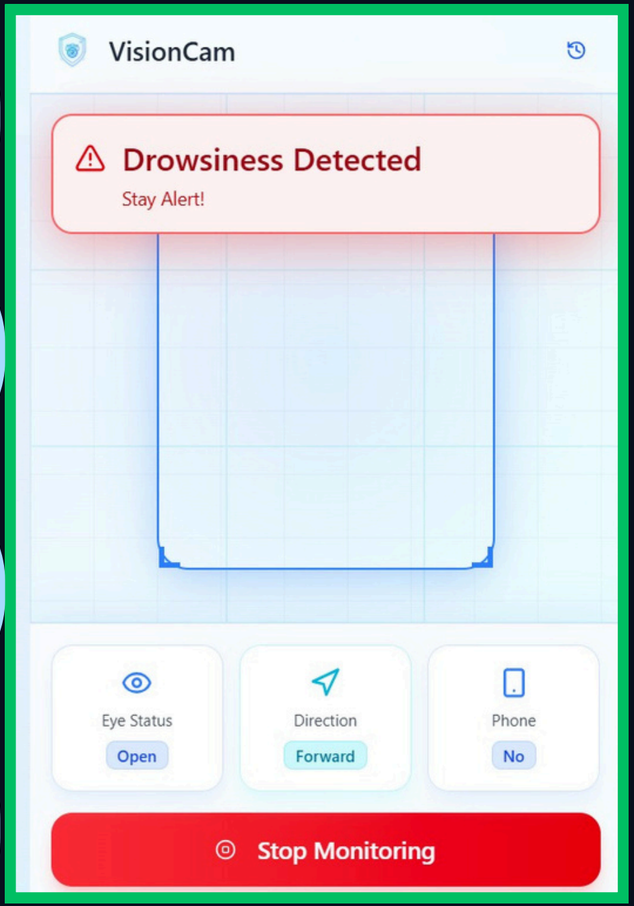
Deployment-

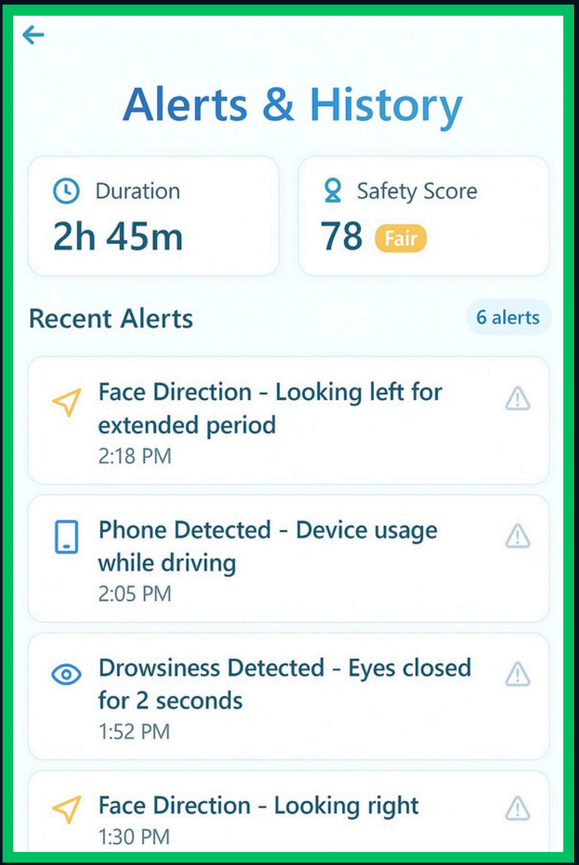
- Vercel / Render (hosted app)
- PWA support (works in mobile browser)



USP

- Works on Any Phone or laptop no expensive hardware
- Real-time alerts for drowsiness, distraction & phone usage
- Regional voice alerts understood by every driver
- Offline mode possible for low-network highways
- Driver Behaviour Analytics – Tracks distraction/drowsiness patterns and gives personalized safety feedback.





INNOVATION AND FEASIBILITY

- AI tracks eye-blink rate, yawning & distraction in real time
- Auto Emergency Call to family/helpline if driver doesn't respond
- Violation logs with time & date stored for safety monitoring
- Non-intrusive real-time monitoring without interrupting or distracting the driver.
- Works even at night using low-light detection

Technical Feasibility: Can run on standard smartphones or in-car systems using lightweight AI/ML for real-time detection.

Practical Feasibility: Scalable, offline-capable, and user-friendly for drivers, taxis, and logistics fleets.

Economic Feasibility: Low-cost development with no extra hardware, deployable as a freemium app or fleet integration.

IMPACT

Enhances Road Safety:
Reduces accidents caused by drowsy or distracted driving.

Saves Lives:
Provides timely alerts that help drivers regain attention and prevent fatal crashes.

Affordable & Accessible:
Works on any smartphone — no expensive hardware required.

Promotes Responsible Driving:
Analytics and feedback help drivers build safer habits.

Supports Public Safety Goals:
Aligns with national and global road safety initiatives (e.g., Vision Zero, WHO Road Safety Targets).

FUTURE SCOPE

Fleet Mode :
Real-time monitoring of multiple drivers for logistics/cab companies.

Wearable Integration :
Detect fatigue accurately using smartwatch/fitness band data.

Cloud Dashboard :
Long-term driving analytics for companies and individuals.

Vehicle Integration –:
Seamless alerts and emergency mode via Bluetooth/infotainment systems.

Automaker Collaboration :
Can be embedded into in-car safety/ADAS systems.